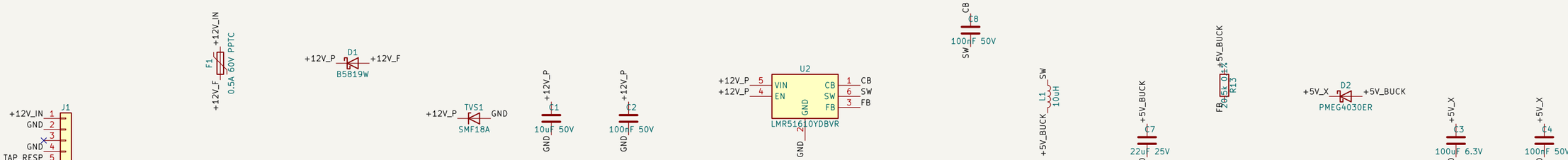
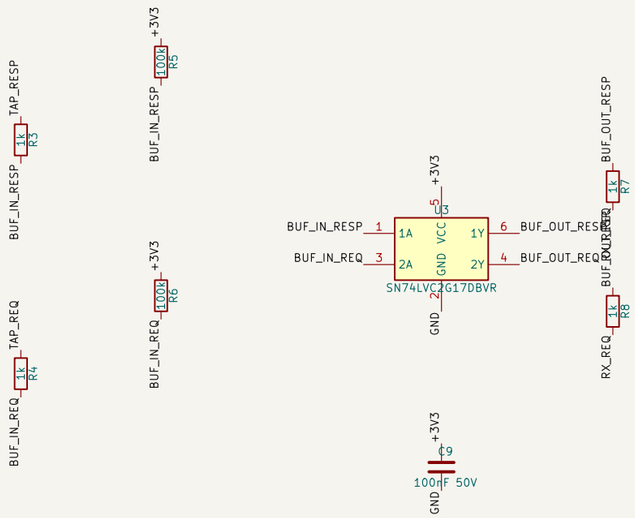
POWER: harness 12 V -> F1 PPTC -> D1 reverse-polarity -> TVS1/C1/C2 -> U2 buck -> D2 backfeed block -> C3/C4 -> XIAO 5V



TAPS (RECEIVE ONLY): pad -> off-board 10k -> J1 -> R3/R4 1k -> U3 Schmitt buffer inputs -> R7/R8 1k -> XIAO D0/D1.

U3 is a one-way barrier: no XIAO pin has a DC path to a tap line. R5/R6 100k bias the buffer inputs high (UART idle) when J1 is off.



DNP: proxy only. R12 + JP1 (open) + TP1 reserve a future GPIO8 TX path; TP1 is NOT wired to J1.

Fitting R12 and bridging JP1 requires an explicitly approved proxy-mode change.



J1 OFF FOR USB: unplug the harness before connecting a plain USB cable (XIAO 5V is fed from the inverter through D2).

J1 wire pads: 1 +12V, 2 GND (power), 3 spare (NC), 4 GND (signal), 5 TAP pad 9 (inverter replies), 6 TAP pad 10 (dongle requests).

Passive Modbus tap carrier for XIAO ESP32S3 Plus (generated by scripts/gen_sch.py)

SolArk_Live_Data

Sheet: /

File: carrier.kicad_sch

Title: SolArk XIAO tap carrier

Size: A3	Date: 2026-10-13	**Rev: B-diy**
KiCad E.D.A. 10.0.6		Id: 1/1